

ENTERPRISE ARCHITECTURE SHOWCASE

# Advanced OCR-To-RAG Retrieval Engine

A Unified Pipeline Combining Adaptive Chunking, OCR-Table Extraction, Metadata Boosting, and Parent-Child Context Restructuring.

 Ingestion Flow

 Hybrid Retrieval

 Metadata-Reranked



PART ONE

# The Core Ingestion Strategy

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Analyzing how documents are split into cohesive units at indexing time using specialized parsing engines.



# Three Ingestion Pillars



## 1. Adaptive Chunking

Splits narrative text purely by paragraph and section boundaries instead of strict word counts. Ensures technical prose retains its context.



## 2. Table-Aware Extraction

Creates explicit child chunks directly from structured OCR rows and table headers. Bypasses standard textual noise.



## 3. Parent-Child Mapping

Establishes precise links where small child chunks act as semantic retrieval points, referencing broader sliding-window parent contexts.



# | Adaptive Chunking Strategy

## Preserving Narrative Coherence

Standard fixed-size chunking strategies systematically cut off technical, regulatory, or financial clauses in mid-sentence, introducing critical gaps in LLM comprehension.

-  **Boundary-Aware:** Splitting aligns with natural header changes.
-  **Overlap Integrity:** Maintains surrounding contextual buffers.
-  **Fallback Safety:** Falls back safely to token thresholds when needed.

Child Chunk (Text)

### Section: Fee Policy

Base service fee applies by account tier. See schedule below for details.

Adaptive Guardrail Active

No orphan sentences allowed. Paragraph integrity is strictly locked.



# Table-Aware OCR Handling

## Unlocking Tabular Structure

Dense tables are highly prone to semantic corruption when processed line-by-line as unstructured text. Our pipeline enforces strict row-to-header preservation.

- ✓ **Primary Row Mapping:** Extracted JSON converts row-level blocks into dedicated child chunks.
- ☒ **Header Inclusion:** Every child row stores the structural header schema within its search footprint.
- ⚠ **Fallback Detection:** Plain text table heuristics take over automatically if OCR JSON fails.





# Parent-Child Chunking Logic

## Retrieval Units (Child)

Child chunks are tiny, targeted snippets (e.g., individual paragraphs, exact OCR rows). Keeping them small minimizes retrieval noise and drastically improves vector math match rates.

Targeted Semantic Density

## Generation Units (Parent)

Parent chunks represent wider sliding-window groups of sequential child chunks. When a child matches, the broader Parent window is passed to the LLM to reconstruct context.

Prevents LLM Context Loss



# Rich Ingestion Metadata

To enhance lexical routing, the pipeline tags each generated child chunk with rich structural metadata:

- 📌 **section\_title:** Allows structural mapping matching query section scopes (e.g., "Fee Policy").
- 📁 **chunk\_type:** Tracks structural role (narrative text, general table, or structured ocr\_table\_row).
- ✓ **is\_table\_like:** True-false flag enabling quick algorithmic boosting for tabular queries.
- 📊 **numeric\_density:** Tracks ratio of numbers, preventing text matches overriding quantitative rows.
- H **table\_header\_tokens:** Connects columns to values explicitly for perfect lookups.
- 📈 **ocr\_confidence:** Slightly down-ranks low-confidence scans to avoid Hallucination inputs.



# | Re-ranking & Metadata Boosting

## The Combined Rerank Equation

To ensure exact numerical lookups don't get buried by general semantic similarities, candidate nodes are evaluated with a normalized score aggregation:

$$S_{\text{final}} = w_{\text{semantic}} \cdot S_{\text{embed}} + w_{\text{metadata}} \cdot S_{\text{meta}} + w_{\text{lexical}} \cdot S_{\text{lex}}$$



| Dynamic Boosting Weights |                        |
|--------------------------|------------------------|
| Query Table Intent       | + Table Weight Boost   |
| Exact Number Match       | + Lexical Weight Boost |
| Header Overlap           | + Section Match Boost  |



# Concrete Chunking Example

| Chunk ID | Chunk Type / Role       | Sample Content Extract                                               | Metadata Profile                          |
|----------|-------------------------|----------------------------------------------------------------------|-------------------------------------------|
| Chunk 1  | Narrative Block (Child) | "Fees are charged by tier. Refer to Fee Schedule for exact rates..." | chunk_type: text<br>section: Fees         |
| Chunk 2  | OCR Table Row (Child)   | "Tier A   \$0 - \$10,000   Fee Rate: 1.50%"                          | chunk_type: ocr_table_row<br>row_index: 0 |
| Chunk 3  | OCR Table Row (Child)   | "Tier B   \$10,010 - \$50,000   Fee Rate: 1.20%"                     | chunk_type: ocr_table_row<br>row_index: 1 |
| Chunk 4  | OCR Table Row (Child)   | "Tier C   \$50,001+   Fee Rate: 0.95%"                               | chunk_type: ocr_table_row<br>row_index: 2 |



# The End-to-End Flow

-  **1. OCR Model Output:** Parsing outputs raw text alongside structured tables.
-  **2. Multi-Path Chunking:** Parallel pipelines build child nodes and parent sliding windows.
-  **3. Combined Search:** Retrieval searches children vectors, reranks via metadata, and serves parental packages.

*Right: System visual mapping demonstrating the unified parsing, windowing and boosting mechanisms.*

600 × 400



# | Why This Architecture Wins

## Semantic Intent Routing

By using metadata re-ranking alongside embeddings, queries searching for qualitative numbers or dense conditions are routed straight to exact data cells instead of losing accuracy.

## Uncompromised Generation

Parent chunk reconstruction ensures the synthesizer (LLM) gets complete, cohesive peripheral descriptions. This eradicates standard document-chunk context dropouts.



# | System Ingestion Controls

| Environment Variable    | Purpose / Function                            |
|-------------------------|-----------------------------------------------|
| TABLE_SPLIT_ROWS        | Sets size limits on OCR structural rows.      |
| PARENT_CHUNK_CHILDREN   | Controls sizing of parent slide windows.      |
| METADATA_NUMERIC_WEIGHT | Tunes boosting strength on math lookup terms. |

## Ready for Production

Our ingestion pipelines natively support high-accuracy parsing over scanned reports, PDFs, and legal contracts with OCR heavy workloads.

[🔗 Adjust environment variables to tune accuracy.](#)



# | Image Sources



<https://weaviate.io/assets/images/hero-372a1930c6d24f89e7d7189207c42125.png>

Source: [weaviate.io](https://weaviate.io)